1. Give definitions to polymer, monomer and oligomer, respectively.
2. Calculate M_n (number average molecular weight), M_w (weight average molecular weight) and the polydispersity index (PDI) for a polymer sample that contains the following mixture.

Mole fraction	Molecular weight
0.1	1000000
0.2	700000
0.4	500000
0.2	250000
0.1	100000

3. List two polymers synthesized by step-growth polymerization and the other two synthesized by chain polymerization.
4. The four fundamental steps of free radical polymerization?
5. Using Carother’s equation to explain: Why is an extreme high extent of conversion required to obtain high molecular weights through a step-growth polymerization?
6. If a chain polymerization system yields a molecular weight lower than needed, what can you do to improve the molecular weight.
7. The features of an anionic polymerization?

Chapter V

1. Concept:

- Polymer
- Monomer
- Oligomer
- Addition polymer vs condensation polymer
- Step-growth polymerization vs chain polymerization
- Number-average/weight-average molecular weight
- Polydispersity PDI
- Thermoplastic and thermoset
- Configuration vs conformation
- Tacticity: atactic; isotactic; syndiotactic

2. Calculation of molecular weights of polymers and PDI

3. Using polypropylene as an example to explain the influence of tacticity on the physical properties of the polymers. (crystallinity; strength; melting temperature and so on)

4. Discriminate between chain polymerization and step-growth polymerization.

5. Step-growth polymerization

- Examples: polyester, polyamide (acid chloride or transesterification involving a carboxylic acid ester); polybenzimidazole; polyimide. (Nylon; PET; PC)
- Carothers equation: the relationship between molecular weights and the extent of reaction (p); and the relationship between molecular weights and the stoichiometric ratio (r); calculate the extent of reaction (p_c) at the gel point.

6. Chain polymerization

- Four fundamental steps: initiation; propagation; termination; chain transfer
- Examples: unsaturated monomers (styrene; methyl methacrylate; acrylonitrile; 1,3-butadinen; vinyl chloride; tetrafluoroethylene; vinyl acetate)
- Can determine a monomer is amenable to a cationic polymerization or an anionic polymerization.
- Free radical polymerization
  The relationship between polymerization rate and temperature.
  The relationship between polymerization rate and activation energy.
  The relationship between molecular weight and initiator concentration/termination/chain transfer.
- Anionic polymerization (a kind of living polymerization): quick initiation; slow propagation; no termination

$$\overline{X_n} = \frac{p[M]_0}{[C]}$$